American Computer Science League

2022-2023 ● Contest 4: Short Problems Solutions ● Intermediate Division

1. Graph Theory

$$\begin{vmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{vmatrix}^2 = \begin{vmatrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{vmatrix}$$

Summing the entries in the squared matrix gives the number of paths of length 2 which is 18.

D. 18

2. Graph Theory

The cycles not containing A are: BCB, BCEDB, and CEDC

B. 3

3. Digital Electronics

The Boolean expression represented by the digital circuit is:

$$\overline{(A + (\overline{AB} + C)) + C}$$

$$\overline{(A + (\overline{AB} + C)) + C} = \overline{A + \overline{A} + \overline{B} + C + C}$$

$$= \overline{1 + \overline{B} + C} = \overline{1} = 0$$

A. 0

4. Digital Electronics

The Boolean expression represented by the digital circuit is:

$$\overline{\overline{A}B\,\overline{(B \oplus C)C}}$$

$$\begin{aligned}\overline{\overline{A}B\,\overline{(B \oplus C)C}} &= \overline{\overline{A}} + \overline{B} + \overline{\overline{(B \oplus C)C}} \\ &= A + \overline{B} + (B \oplus C)C \\ &= A + \overline{B} + (B\overline{C} + \overline{B}C)C \\ &= A + \overline{B} + 0 + \overline{B}C \\ &= A + \overline{B}(1 + C) \\ &= A + \overline{B}\end{aligned}$$

If $A + \overline{B} = 0$, then $A = 0$, $\overline{B} = 0$, $C = *$
Therefore there are 2 ordered tuples that make it FALSE: (0, 1, *)

B. 2

5. Assembly Language

It takes twelve times through the loop for a power of 2 to be larger than a multiple of 100. This is the Payroll Problem of $100 per day versus $1 the first day and doubled each day after that, but any power and any multiple could be used.

C. 12